Distributionally Robust LMI Control

Data-Driven and Distributionally Robust Controller Synthesis

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Chapter 1

Distributionally Robust LMI Synthesis for LTI Systems

1.1 Problem Setting

Distributionally Robust Control (DRC) addresses stochastic systems whose disturbance distribution is only approximately known. Let the plant be

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix}, \qquad \begin{bmatrix} x_{c,t+1} \\ u_t \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} x_{c,t} \\ y_t \end{bmatrix}. \tag{1.1}$$

The closed loop has combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and matrices

$$\mathbb{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \ \mathbb{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix}, \ \mathbb{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}. \tag{1.2}$$

1.2 Ambiguity Sets and Objective

The disturbance w_t is random with unknown law P_w. Instead of assuming a fixed Gaussian, the controller minimizes the *worst-case expected cost* over all distributions inside a 2-Wasserstein ball

$$W_2(P_w, P_{\text{nom}}) \le \gamma, \qquad P_{\text{nom}} = \mathcal{N}(0, \Sigma_{\text{nom}}).$$

Two ambiguity models are treated: temporally correlated ($\mathcal{W}_{\text{cor}}$) and independent ($\mathcal{W}_{\text{ind}}$).

The infinite-horizon quadratic cost reads

$$v(A_c, B_c, C_c, D_c) = \sup_{P_w \in \mathcal{W}} \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|z_t\|^2. \tag{1.3}$$

1.3 Moment Relaxation and LMI Formulation

Gramlich *et al.* showed that the worst-case distribution depends only on second-order moments, yielding an exact moment relaxation. Consequently the DRC problem is equivalent to a robust H_2 synthesis problem and can be written as a finite-dimensional convex program:

$$
\begin{aligned}
\text{minimize} \quad & \operatorname{tr}(Q) && (1.4)\\
\text{subject to} \quad & \begin{bmatrix} \mathbb{A}^\top P \mathbb{A} - P & \mathbb{A}^\top P \mathbb{B} \\ \mathbb{B}^\top P \mathbb{A} & \mathbb{B}^\top P \mathbb{B} - \lambda I \end{bmatrix} \prec 0, && (1.5)\\
& P \succ 0,\ \lambda > 0,\ W_2(\Sigma, \Sigma_{\text{nom}}) \leq \gamma. && (1.6)
\end{aligned}
$$

The resulting semidefinite program produces the distributionally-robust controller parameters (A_c, B_c, C_c, D_c) with the same computational complexity as classical H_2 LMI synthesis, but with statistical robustness to data uncertainty.

Chapter 2

Repository Structure and Implementation

2.1 Overview

This repository implements the framework of *Distributionally Robust LMI Synthesis for LTI Systems* and integrates techniques from *Bias-Correction and Instrumental Variables for Direct Data-Driven Control* and *Data-Enabled Policy Optimization for Direct Adaptive Learning of the LQR*. The goal is a fully data-driven pipeline capable of

1. estimating plant matrices $(A, B_w, B_u, C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ directly from noisy data,
2. constructing the distributionally-robust LMI (DRO-LMI) problem,
3. solving it via convex optimization,
4. reconstructing the optimal controller and validating it in closed-loop simulations.

2.2 Code Architecture

- **main.py** — entry point defining two pipelines:
  1. `baseline_optim_problem`: classical Monte-Carlo cost minimization;
  2. `lmi_pipeline_optim_problem`: distributionally robust LMI synthesis.
- **problem__baseline.py** — stochastic simulation-based optimization estimating $\mathbb{E}[\|z_t\|^2]$ by long-run averaging; includes stability projection heuristics.
- **problem__dro_lmi.py** — builds the DRO-LMI following the Wasserstein formulations for correlated or independent disturbances and solves it with CVXPY.
- **utils__ambiguity.py** — definition of the 2-Wasserstein ambiguity set and projection of covariance matrices onto the admissible ball $B(\Sigma_{\text{nom}}, \gamma)$.
- **utils__matrices.py** — symbolic composition and recovery of composite matrices, reconstruction of controller matrices from LMI variables (X, Y, K, L, M, N).
- **utils__simulate.py** — time-domain simulation of the closed-loop system and Monte-Carlo validation under sampled disturbances.

- **utils__systems.py** — data classes for `Plant`, `Controller`, and `DROLMIResult`.

2.3 Mathematical Workflow

2.3.1 1. Plant Identification

Measured sequences $\{x_t, u_t\}$ are aligned to form

$$[X_+, U_0] = \begin{bmatrix} x_1 & \cdots & x_T \\ u_0 & \cdots & u_{T-1} \end{bmatrix}, \qquad [A\ B_u] = X_+ V^\top (VV^\top + \rho I)^{-1},$$

with optional bias-correction (BC) or instrumental variables (IV) to mitigate noise bias, following [1].

2.3.2 2. Ambiguity Set Definition

A nominal covariance Σ_{nom} and radius γ define the Wasserstein ball $\mathcal{W}$. The code supports both independent and correlated models.

2.3.3 3. LMI Formulation

Matrices (X, Y, K, L, M, N) parameterize the closed-loop as in [2]. Convex constraints enforce Lyapunov stability and bounded cost for all disturbance distributions in $\mathcal{W}$.

2.3.4 4. Controller Reconstruction

From the optimal (X, Y, K, L, M, N), the physical controller matrices are recovered via

$$\begin{aligned} A_c &= U^{-1}\big(K - XAY - LC_yY - XB_u(M - NC_yY)\big)V^{-\top}, \\ B_c &= U^{-1}(L - XB_uN), \quad C_c = (M - NC_yY)V^{-\top}, \quad D_c = N. \end{aligned}$$

2.3.5 5. Closed-Loop Validation

The resulting controller is tested using Monte-Carlo simulations with disturbances sampled from $\mathcal{N}(0, \Sigma)$, both nominal and adversarial, to evaluate average cost and stability margins.

2.4 Relation to the Reference Papers

- From [2]: convex DRO-LMI formulation and equivalence to robust-H_2 synthesis.
- From [3]: covariance-based parameterization ensuring data-length-independent decision size.
- From [1]: bias-correction and instrumental-variable schemes for noise mitigation in direct data-driven identification.

2.5 Outcome

The repository reproduces a complete distributionally-robust data-driven controller design chain:

Data $\rightarrow$ Matrix Estm $\rightarrow$ DRO-LMI Opt $\rightarrow$ Controller Recovery $\rightarrow$ Sim and Comp.

It provides a reproducible platform for testing robust-H_2 and direct data-driven synthesis methods under Wasserstein-bounded uncertainty.

Chapter 3

Optimisation Pipelines: Pseudocode and Mathematical Steps

3.1 Notation and Closed-Loop Composition

Let the plant be

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix}, \qquad \begin{bmatrix} x_{c,t+1} \\ u_t \end{bmatrix} = \begin{bmatrix} A_c & B_c \\ C_c & D_c \end{bmatrix} \begin{bmatrix} x_{c,t} \\ y_t \end{bmatrix}.$$

Define the combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and the closed-loop matrices

$$\mathbb{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathbb{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathbb{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathbb{D} = D_{zw} + D_{zu} D_c D_{yw}.$$

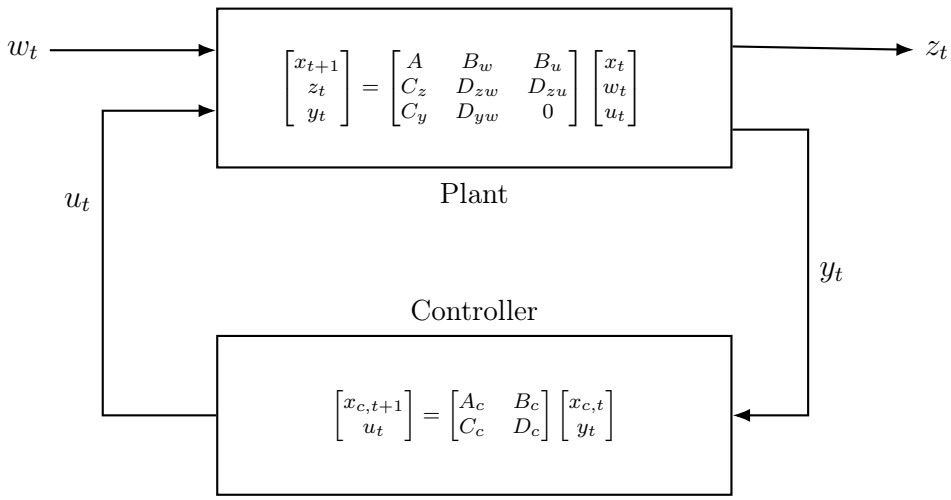

Figure 3.1: Closed-loop interconnection with controller placed below the plant.

$$\chi_t = \begin{bmatrix} x_t \\ x_{c,t} \end{bmatrix}$$

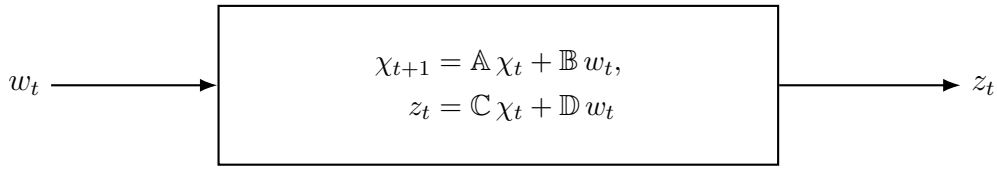

Figure 3.2: Lifted closed loop with state χ_t and matrices $(\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D})$.

3.2 Baseline Monte-Carlo Cost Minimization (State-Feedback Controller Search)

We minimize the long-run average quadratic performance

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathbb{A} \quad \text{isstable}$$

$$J(\theta) \;=\; \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\, \|z_t\|^2$$

where θ stacks the entries of (A_c, B_c, C_c, D_c), $w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}})$. A stability penalty is added if $\rho(\mathbb{A}) \geq 1$.

Algorithm 1 Baseline Optimization via Monte-Carlo Cost

Require: Plant $(A, B_w, B_u, C_z, D_{zw}, D_{zu}, C_y, D_{yw})$, initial controller $(A_c^0, B_c^0, C_c^0, D_c^0)$, covariance $\Sigma_{\text{nom}} \succ 0$, seeds $\mathcal{S}$, horizon T, burn-in T_b

1: Parameterize $\theta \leftrightarrow (A_c, B_c, C_c, D_c)$ with simple box bounds.
2: Define stability test stable$(\mathbb{A}) \Leftrightarrow \rho(\mathbb{A}) < 1$.
3: Define penalized objective

$$\tilde{J}(\theta) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \left[\frac{1}{T - T_b} \sum_{t=T_b}^{T-1} \|z_t^{(s)}\|^2 \right] \;+\; \lambda_\rho \, \max\{0, \rho(\mathbb{A}(\theta)) - 1\}^2.$$

4: For each seed $s \in \mathcal{S}$: simulate

$$\begin{aligned} y_t &= C_y x_t + D_{yw} w_t, \qquad u_t = C_c x_{c,t} + D_c y_t, \\ z_t &= C_z x_t + D_{zu} u_t + D_{zw} w_t, \\ x_{t+1} &= A x_t + B_u u_t + B_w w_t, \qquad x_{c,t+1} = A_c x_{c,t} + B_c y_t, \end{aligned}$$

with $w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}})$.
5: Use L-BFGS-B (or similar) to compute $\theta^\star = \arg\min_\theta \tilde{J}(\theta)$.
6: If $\rho(\mathbb{A}(\theta^\star)) \geq 1$, apply a stabilizing projection to (A_c, D_c) and re-evaluate.
7: **return** optimized $(A_c^\star, B_c^\star, C_c^\star, D_c^\star)$ and the estimated cost.

Remarks. (i) The estimator of J is a single-trajectory long-run average after burn-in. (ii) The penalty keeps the search in a practically stable region. (iii) Bounds on θ curb numerical blow-ups.

3.3 Distributionally Robust LMI Synthesis (Wasserstein Ambiguity)

We seek a controller that minimizes the worst-case stationary quadratic cost over the 2-Wasserstein ball around a nominal Gaussian:

$$\min_{A_c,B_c,C_c,D_c} \sup_{P_w:\, W_2(P_w,\mathcal{N}(0,\Sigma_{\text{nom}}))\le\gamma} \limsup_{T\to\infty} \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\|z_t\|^2.$$

Using a moment relaxation that is tight for this setting, introduce decision variables

$$\lambda \ge 0, \quad Q \succeq 0, \quad X = X^\top,\ Y = Y^\top,\ K, L, M, N,$$

and define

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

The objective is

$$\min_{\lambda\ge 0,\ X=X^\top,\ Y=Y^\top,\ K,L,M,N} \operatorname{trace}(Q\Sigma_{\text{nom}}) + \lambda\,\gamma^2$$

subject to a block LMI that depends on the ambiguity model.

Correlated model ($\mathcal{W}_{\text{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \qquad Q \succeq 0, \quad \mathbb{P} \succeq 0.$$

Independent model ($\mathcal{W}_{\text{ind}}$). Impose the pair of LMIs

$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad Q \succeq 0,\ \mathbb{P} \succeq 0.$$

Algorithm 2 DRO-LMI Synthesis under Wasserstein Ambiguity

Require: Plant $(A, B_w, B_u, C_z, D_{zw}, D_{zu}, C_y, D_{yw})$, nominal covariance $\Sigma_{\text{nom}} \succ 0$, radius $\gamma \geq 0$, model $\in \{\text{correlated}, \text{independent}\}$

1: Introduce variables $\lambda, Q, X, Y, K, L, M, N$ and build $\mathbb{P}, \mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ as above.
2: Formulate the model-specific LMI constraints (correlated or independent).
3: Minimize $\text{trace}(Q\Sigma_{\text{nom}}) + \lambda\gamma^2$ subject to the LMIs.
4: Solve the SDP with a conic solver (e.g., MOSEK/CVXOPT/SCS).
5: Recover a controller by mapping $(X, Y, K, L, M, N) \mapsto (A_c, B_c, C_c, D_c)$.
6: Validate closed-loop stability and evaluate performance with Monte-Carlo tests.
7: **return** (A_c, B_c, C_c, D_c), objective value, and diagnostic spectra.

Controller recovery. Given optimal (X, Y, K, L, M, N), compute a realization (A_c, B_c, C_c, D_c) consistent with the closed-loop parameterization. One convenient form is

$$D_c = N, \qquad B_c \approx (L - XB_uN)_{\text{factored}}, \qquad C_c \approx (M - NC_yY)_{\text{factored}},$$

and

$$A_c \approx K - XAY - LC_yY - XB_u\,(M - NC_yY),$$

optionally using a numerically well-conditioned factorization of $(I - XY)$ to ensure realizability.

3.4 Coordinate Transformation in the LMI Formulation

Following [2], the LMI constraints are written in a transformed coordinate system defined by a nonsingular matrix T. The closed-loop matrices $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ are linked to the physical realization (A, B, C, D) through

$$\begin{pmatrix} T^\top \mathcal{P} A T & T^\top \mathcal{P} B \\ CT & D \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}, \qquad T^\top \mathcal{P} T = P. \tag{3.1}$$

The matrix $\mathcal{P}$ is the lifted Lyapunov variable

$$\mathcal{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad P = \mathcal{P}(X, Y),$$

appearing in the block LMI that enforces the distributionally robust H_2 performance condition.

Choice of T. In principle, one could pick T such that it is orthogonal with respect to $\mathcal{P}$, i.e. $T^\top \mathcal{P} T = I$, which simplifies the Lyapunov constraints. However, $\mathcal{P}$ is itself an optimization variable, and enforcing orthogonality between T and $\mathcal{P}$ would make the problem bilinear and therefore *nonconvex*. To preserve convexity of the LMI problem, we fix

$$T = I_{2n_x},$$

so that $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ coincide with the matrices of the original lifted closed loop. This choice leaves the synthesis convex and allows standard SDP solvers to handle the problem efficiently.

Remark. Under this choice, the equivalence between the distributionally robust formulation and the standard robust-H_2 LMI synthesis remains exact, as the transformation acts only as a coordinate scaling in the Lyapunov space.

3.5 Remark on the Transformation Matrix T

In the LMI formulation, one may introduce a transformation T such that

$$\mathbf{A} = T^\top A T, \quad \mathbf{B} = T^\top B, \quad \mathbf{C} = CT.$$

Ideally, T should satisfy the orthogonality condition $T^\top P T = I$, making it aligned with the Lyapunov variable P. However, since P is itself an optimization variable, enforcing this constraint would destroy convexity. Therefore, in this implementation we set $T = I$, which preserves convexity of the LMI problem while maintaining interpretability of the lifted matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$.

3.6 System Definition and Data-Driven Recovery

3.6.1 Paper-like Model Setting

The *paper-like* configuration refers to a linear time-invariant (LTI) system of the form:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ denotes the state, $u_t \in \mathbb{R}^{n_u}$ the control input, and $w_t \in \mathbb{R}^{n_w}$ an exogenous disturbance. The vectors $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ represent the measured and performance outputs, respectively.

The continuous-time version of this model describes a flexible turbine–shaft–pitch system:

$$\dot{x}(t) = A_c x(t) + B_c u(t) + E_c w(t),$$

with

$$A_c = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_c = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

which is discretized with sampling time T_s via

$$A = e^{A_c T_s}, \qquad B_u = \int_0^{T_s} e^{A_c \tau} B_c \, d\tau, \qquad B_w = \int_0^{T_s} e^{A_c \tau} E_c \, d\tau.$$

The output matrices $(C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ are selected to reflect a standard $\mathcal{H}_2$-type performance objective:

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0 \qquad y_t = C_y x_t + D_{yw} w_t,$$

with $Q_x \succeq 0, R_u \succ 0$. When the configuration parameter `outputs.mode = A`, this corresponds to a *state-input performance formulation* where $D_{zw} = 0$.

3.7 Data-Driven Recovery of the Paper-Like Matrices

We estimate the plant

$$\mathcal{P} = (A, B_u, B_w, C_z, D_{zw}, D_{zu}, C_y, D_{yw})$$

directly from a single trajectory $\{x_t, u_t, y_t, z_t\}_{t=0}^{T-1}$ by aligned one-step regression. Throughout, vectors are column-valued, all sample matrices are column-stacked (variables by time), and expectations are replaced by empirical averages (with ridge terms to tame conditioning).

3.7.1 Preprocessing and Notation

Blocks. Build aligned blocks over a common horizon $T-1$:

$$\begin{aligned}
&X \in \mathbb{R}^{n_x \times (T-1)} && X := [\, x_0 \; \dots \; x_{T-2} \,], \\
&U \in \mathbb{R}^{n_u \times (T-1)} && U := [\, u_0 \; \dots \; u_{T-2} \,], \\
&X^+ \in \mathbb{R}^{n_x \times (T-1)} && X^+ := [\, x_1 \; \dots \; x_{T-1} \,], \\
&Y \in \mathbb{R}^{n_y \times (T-1)}, && Z \in \mathbb{R}^{n_z \times (T-1)} \text{ if available.}
\end{aligned}$$

Centering. Work with demeaned data (rowwise):

$$\tilde{X} := X - \bar{x}\,\mathbb{1}^\top, \quad \tilde{U} := U - \bar{u}\,\mathbb{1}^\top, \quad \tilde{X}^+ := X^+ - \bar{x}^+\,\mathbb{1}^\top,$$

and define the concatenated regressor $D := \begin{bmatrix} \tilde{X} \\ \tilde{U} \end{bmatrix} \in \mathbb{R}^{(n_x+n_u)\times(T-1)}$.

Persistency. We require the empirical persistency-of-excitation (PE) condition

$$rk(D) = n_x + n_u,$$

else the identification is ill-posed.

3.7.2 State and Input Matrices (A, B_u)

Ridge-regularized least squares gives the minimum-variance linear predictor for x_{t+1} given (x_t, u_t):

$$\boxed{\begin{bmatrix} A & B_u \end{bmatrix} \;=\; \tilde{X}^+ D^\top \left(DD^\top + \lambda I_{n_x+n_u}\right)^{-1},} \qquad \lambda > 0. \tag{3.2}$$

Shapes: $\tilde{X}^+ \in \mathbb{R}^{n_x \times (T-1)}$, $DD^\top \in \mathbb{R}^{(n_x+n_u)\times(n_x+n_u)}$.

3.7.3 Optional Stability Projection (Spectral Shrink)

Why instability appears. Given finite data, the ridge estimator of the state matrix

$$\widehat{A} = \tilde{X}^+ \tilde{X}^\top \left(\tilde{X}\tilde{X}^\top + \lambda I_{n_x}\right)^{-1} \quad \text{or} \quad \widehat{A} = \tilde{X}^+ D^\top \left(DD^\top + \lambda I\right)^{-1}$$

(with $D = [\tilde{X}; \tilde{U}]$) is statistically noisy. Even if the true discrete-time system is stable, sample noise, finite horizon, and numerical conditioning can push some eigenvalues outside the unit disk:

$$\rho(\widehat{A}) = \max_i |\hat{\mu}_i| \geq 1.$$

This is an estimation artifact that makes open-loop simulations diverge.

Spectral decomposition. Assume $\widehat{A}$ is diagonalizable:

$$\widehat{A} = V\,\mathrm{diag}(\hat{\mu}_1, \ldots, \hat{\mu}_{n_x})\,V^{-1},$$

where $\hat{\mu}_i \in \mathbb{C}$ are eigenvalues and V contains the corresponding eigenvectors.

Stability requirement (discrete time). A stable discrete-time model satisfies $|\mu_i| < 1$ for all i. If any $|\hat{\mu}_i| \geq 1$, trajectories grow roughly like $\hat{\mu}_i^t$ along mode v_i.

The projection (what we actually do). Choose a radius $\rho_\star \in (0, 1)$ (e.g. $\rho_\star = 0.995$) and define the shrunk eigenvalues

$$\tilde{\mu}_i \;=\; \begin{cases} \hat{\mu}_i, & |\hat{\mu}_i| < \rho_\star, \\ \rho_\star\,\dfrac{\hat{\mu}_i}{|\hat{\mu}_i|}, & |\hat{\mu}_i| \geq \rho_\star. \end{cases}$$

Recompose

$$A_{\mathrm{proj}} \;=\; V\,\mathrm{diag}(\tilde{\mu}_1, \ldots, \tilde{\mu}_{n_x})\,V^{-1}, \quad \text{so that} \quad \rho(A_{\mathrm{proj}}) \leq \rho_\star < 1.$$

Equivalently, in one line:

$$A_{\mathrm{proj}} = V\,\mathrm{diag}\left(\min\{|\hat{\mu}_i|, \rho_\star\}\,\frac{\hat{\mu}_i}{|\hat{\mu}_i|}\right) V^{-1}.$$

Where each term comes from.

- V and $\hat{\mu}_i$ come from the eigendecomposition of $\widehat{A}$: $\widehat{A} = V\mathrm{diag}(\hat{\mu}_i)V^{-1}$.
- $\rho_\star$ is a user-chosen contraction radius to ensure strict stability.
- $\frac{\hat{\mu}_i}{|\hat{\mu}_i|}$ preserves the phase (argument) of complex eigenvalues, so oscillation frequency is unchanged; only the magnitude is reduced.
- $\min\{|\hat{\mu}_i|, \rho_\star\}$ performs a radial clipping in the complex plane, projecting any offending eigenvalue back inside the disk of radius $\rho_\star$.

Geometric interpretation. The mapping $\hat{\mu}_i \mapsto \rho_\star\,\hat{\mu}_i/|\hat{\mu}_i|$ is a radial projection onto the closed disk $\{z \in \mathbb{C} : \; |z| \leq \rho_\star\}$. Angles are preserved; magnitudes are damped.

Why this is acceptable. The operation preserves mode directions (V) and enforces $\rho(A_{\mathrm{proj}}) < 1$, which:

1. avoids divergence in time-domain simulations,
2. helps feasibility of Lyapunov LMIs of the form $A^\top PA - P \prec 0$,
3. trades a small bias for a large variance reduction in finite-sample identification near the stability boundary.

When to use it. Use it for discrete-time identified models before simulation or LMI synthesis when finite samples yield marginal/unstable $\widehat{A}$. Avoid it if the goal is unbiased eigenvalue inference rather than a stable surrogate model.

Numerical note. If $\widehat{A}$ is defective (non-diagonalizable), replace the eigendecomposition with a Schur decomposition $Q^\top \widehat{A} Q = T$ and shrink the eigenvalues on the (block-)diagonal of T; then set $A_{\text{proj}} = Q T_{\text{shrunk}} Q^\top$ to maintain numerical robustness.

3.7.4 Disturbance Channels B_w from Residuals

Residual construction. Once (A, B_u) are estimated by regression, the remaining unexplained dynamics at each step

$$R := \tilde{X}^+ - A\,\tilde{X} - B_u\,\tilde{U} \in \mathbb{R}^{n_x \times (T-1)}$$

represent the *innovation sequence*—the portion of the next state not predicted by (x_t, u_t).

In the ideal linear-Gaussian model

$$x_{t+1} = Ax_t + B_u u_t + B_w w_t, \qquad w_t \sim \mathcal{N}(0, I_{n_w}),$$

we would have $R \approx B_w W$, where $W = [w_0, \ldots, w_{T-2}]$ is white with identity covariance, so that

$$\frac{1}{T-1} RR^\top \approx B_w B_w^\top.$$

Hence, the residual covariance empirically estimates the disturbance covariance propagated through B_w.

Covariance decomposition. Define the residual covariance matrix

$$S_R := \frac{1}{T-1} RR^\top = U_R\,\Sigma_R\,U_R^\top,$$

with eigenvalues $\Sigma_R = \operatorname{diag}(\sigma_1 \geq \cdots \geq \sigma_{n_x} \geq 0)$. Each σ_i represents the energy of a principal disturbance direction.

Rank truncation and energy criterion. Noise and modeling imperfections make S_R full-rank even if the true B_w has low dimension $n_w < n_x$. To recover a parsimonious model, retain the smallest k such that

$$\frac{\sum_{i=1}^{k} \sigma_i}{\sum_{i=1}^{n_x} \sigma_i} \;\geq\; \eta, \qquad \eta \in (0, 1] \text{ (e.g. } \eta = 0.95).$$

Then construct

$$\boxed{B_w := U_R(:, 1{:}k)\;\Sigma_R(1{:}k, 1{:}k)^{1/2} \;\in\; \mathbb{R}^{n_x \times k},} \qquad n_w := k. \tag{3.3}$$

This ensures $B_w B_w^\top$ reproduces the dominant k-dimensional subspace of S_R and that the retained modes account for a fraction η of the total residual variance.

Interpretation. B_w provides a low-dimensional basis for the "innovation" space: each column of B_w corresponds to one independent disturbance channel driving the state dynamics. The scaling by $\Sigma_R^{1/2}$ gives physically meaningful amplitudes, so that $\mathrm{Cov}(B_w w_t) = B_w B_w^\top \approx S_R$.

Why this approach is effective for estimation.

- **Model-free:** It does not assume any explicit statistical model for w_t—the residuals themselves reveal the disturbance directions supported by the data.
- **Noise separation:** By isolating dominant singular directions of S_R, the method separates structured process noise (which is systematic) from small unstructured measurement noise (which appears in low-energy directions).
- **Dimension reduction:** The energy criterion η yields a parsimonious representation of the disturbance space, preventing overfitting and improving numerical conditioning in subsequent LMI steps.
- **Consistency:** If (A, B_u) are consistent and w_t is i.i.d. Gaussian, then $S_R \to B_w B_w^\top$ in probability as $T \to \infty$, so the estimated B_w converges (up to an orthogonal rotation).
- **Physical interpretability:** Each retained mode corresponds to an independent disturbance direction acting on the state, analogous to empirical principal components of process noise.

Dimensionality and implications for optimization. The procedure above yields $B_w \in \mathbb{R}^{n_x \times k}$, where $k = n_w$ is the estimated disturbance dimension determined by the residual energy threshold. In practice, the identified k may differ from the true disturbance dimension used in the data-generating process. When k is underestimated (e.g., $n_w = 1$ while the true system has higher-rank noise), the identified plant effectively models process disturbances as acting along a single dominant direction. For the baseline optimization this usually remains numerically feasible but can produce optimistic performance estimates, since only a reduced subset of disturbance directions is represented in simulation. In the distributionally robust LMI formulation, however, the ambiguity set is constructed around the nominal covariance $\Sigma_{\mathrm{nom}} = B_w B_w^\top$ and thus inherits its column space: the worst-case perturbations are confined to $\mathrm{span}(B_w)$. If this span is too narrow, the resulting controller is not robust to disturbances outside that subspace. To mitigate this, one can either increase the retained rank k, lower the energy threshold η, or augment B_w with a small isotropic component (e.g. $B_w \leftarrow [\, B_w \;\; \delta I_{n_x} \,]$ with $\delta \ll 1$) so that the ambiguity set meaningfully covers all disturbance directions. This adjustment preserves convexity and ensures that both the baseline and the DRO-LMI synthesis remain well-posed and robust to realistic noise geometries.

Summary. The SVD-based reconstruction of B_w provides a statistically grounded and numerically robust estimate of the process-noise channels. It captures the empirical energy distribution of residual dynamics without requiring explicit noise measurements or second-run experiments.

3.7.5 Output Channels (C_y, D_{yw}) and Performance Map (C_z, D_{zu}, D_{zw})

When Y and/or Z are available, perform ridge regressions that explicitly separate the state pathway and the residual pathway.

Measured output. Model $y_t = C_y x_t + D_{yw} w_t$ with w_t proxied by residuals. Stacking columns:

$$\boxed{Y \approx \underbrace{C_y}_{n_y \times n_x} \tilde{X} \;+\; \underbrace{D_{yw}}_{n_y \times n_w} R_w, \qquad R_w := B_w^{\dagger} R,}$$

and solve the ridge problem

$$\min_{C_y, D_{yw}} \; \big\| Y - [\, C_y \;\; D_{yw} \,]\, [\, \tilde{X};\, R_w \,] \big\|_F^2 \;+\; \lambda_y \| [\, C_y \;\; D_{yw} \,] \|_F^2,$$

which yields the closed form

$$[\, \widehat{C}_y \;\; \widehat{D}_{yw} \,] = Y\, \Theta_Y^{\top} (\Theta_Y \Theta_Y^{\top} + \lambda_y I)^{-1}, \quad \Theta_Y := [\, \tilde{X};\, R_w \,].$$

Performance output. Model $z_t = C_z x_t + D_{zu} u_t + D_{zw} w_t$ with the same proxy:

$$\boxed{Z \approx C_z \tilde{X} + D_{zu} \tilde{U} + D_{zw} R_w,}$$

and solve

$$[\, \widehat{C}_z \;\; \widehat{D}_{zu} \;\; \widehat{D}_{zw} \,] = Z\, \Theta_Z^{\top} (\Theta_Z \Theta_Z^{\top} + \lambda_z I)^{-1}, \quad \Theta_Z := [\, \tilde{X};\, \tilde{U};\, R_w \,].$$

3.7.6 Bias-Correction (BC) and Instrumental Variables (IV) Options

Noise in x_t biases (3.2). Two classical mitigations are supported.

BC (measurement-noise correction). Let v_t be i.i.d. sensor noise in x_t with variance proxy σ_v^2 (estimated e.g. by first differences). In the moment domain,

$$\underbrace{\frac{1}{T-1} X \Phi^{\top}}_{\bar{X}_0} \;\approx\; \underbrace{\frac{1}{T-1} X^{+} \Phi^{\top}}_{\bar{X}_1} \big[A \;\; B_u \big]^{\top} \;+\; \Psi, \qquad \Phi := \begin{bmatrix} U \\ X \end{bmatrix},$$

where $\Psi = \big[0_{n_x \times n_u} \;\; (T-1)\sigma_v^2 I_{n_x} \big]$ accounts for the noise bias. Replace $\bar{X}_0 \leftarrow \bar{X}_0 - \Psi/(T-1)$ before solving the normal equations. This preserves convexity; accuracy hinges on the σ_v^2 proxy.

IV (two-run instruments). Given an instrument sequence X^{iv} with the same input U but independent measurement noise,

$$\min_{A, B_u} \; \big\| \tilde{X}^{+} \Phi_{\mathrm{iv}}^{\top} - [A \;\; B_u]\, D_{\mathrm{iv}} \big\|_F^2, \quad \Phi_{\mathrm{iv}} := \begin{bmatrix} U \\ X^{\mathrm{iv}} \end{bmatrix}, \quad D_{\mathrm{iv}} := \begin{bmatrix} \tilde{X} \\ \tilde{U} \end{bmatrix},$$

leading to

$$[\, \widehat{A} \;\; \widehat{B}_u \,] = \tilde{X}^{+} \Phi_{\mathrm{iv}}^{\top} \big(D_{\mathrm{iv}} \Phi_{\mathrm{iv}}^{\top} \big)^{\dagger}.$$

Rank conditions now apply to D_{iv} and Φ_{iv}; numerically, ridge regularization is still recommended.

3.7.7 Putting It Together

The identified plant is

$$\widehat{\mathcal{P}} = (\widehat{A}, \widehat{B}_u, \widehat{B}_w, \widehat{C}_z, \widehat{D}_{zw}, \widehat{D}_{zu}, \widehat{C}_y, \widehat{D}_{yw}),$$

with $\widehat{B}_w$ from (3.3), and outputs from the ridge problems above.

3.7.8 Diagnostics (what you should actually check)

- **PE/conditioning:** $(D) = n_x + n_u$ and $\kappa(DD^\top + \lambda I)$ is moderate.
- **Dynamic fit:** relative residual $\|R\|_F / \|\tilde{X}^+\|_F$.
- **Model order for B_w:** variance energy captured $\sum_{i=1}^{k} \sigma_i / \sum_i \sigma_i$.
- **Stability:** $\rho(\widehat{A}) < 1$ or post-projection radius < 1 if you applied shrinkage.

Assumptions and limitations. All regressions above assume ergodicity and that centering removes mean offsets; BC quality depends on the σ_v^2 proxy; IV needs an instrument sequence with independent sensor noise; output regressions treat R_w as a proxy for w_t, which is exact only in the ideal linear-Gaussian limit with perfect $B_w^\dagger$.

3.8 Disturbance Sampling for Simulation and Robust Evaluation

The disturbance sequences w_t used in both the baseline and distributionally robust simulations are drawn from a *Wasserstein ambiguity set* centered at a nominal Gaussian model. The ambiguity set captures distributional uncertainty in the second-order statistics of the process noise, rather than assuming a fixed covariance.

Definition of the ambiguity set. For a given nominal covariance $\Sigma_{\text{nom}} \succ 0$ and Wasserstein radius $\gamma > 0$, the 2-Wasserstein ambiguity set is defined as

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \Big\{ \mathcal{N}(0, \Sigma) \;\Big|\; W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma \Big\},$$

where $W_2(\cdot, \cdot)$ denotes the 2-Wasserstein distance between Gaussian measures:

$$W_2^2(\mathcal{N}(0, S_1), \mathcal{N}(0, S_2)) = \operatorname{tr}(S_1) + \operatorname{tr}(S_2) - 2 \operatorname{tr}\Big[\big(S_2^{1/2} S_1 S_2^{1/2} \big)^{1/2} \Big].$$

This set represents all Gaussian distributions whose covariance lies within distance γ (in the Bures metric) from the nominal covariance Σ_{nom}.

Sampling strategies. Two simulation modes are available:

1. **Independent (iid) disturbances:**

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}}), \quad t = 0, \ldots, T-1,$$

   corresponding to temporally uncorrelated samples inside the ambiguity ball. Implementation uses a Cholesky factorization $L = \operatorname{chol}(\Sigma_{\text{nom}})$ and generates $w_t = L\, z_t,\ z_t \sim \mathcal{N}(0, I)$.

2. **Correlated disturbances:** To allow temporal correlation while preserving the same marginal covariance, an autoregressive model of order one (AR(1)) is simulated:

$$w_{t+1} = \rho\, w_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

   with $0 < \rho < 1$ ensuring stability. The matrix Q is chosen so that

$$\Sigma_{\text{nom}} = \rho^2 \Sigma_{\text{nom}} + Q \quad \Rightarrow \quad Q = (1 - \rho^2)\Sigma_{\text{nom}}.$$

   This yields stationary correlated samples having marginal covariance Σ_{nom}, ensuring consistency with the ambiguity model.

Integration in the simulation. Each disturbance sequence $w = \{w_t\}_{t=0}^{T-1}$ is generated before the closed-loop simulation and injected into the identified plant dynamics as

$$x_{t+1} = Ax_t + B_u u_t + B_w w_t, \qquad z_t = C_z x_t + D_{zu} u_t + D_{zw} w_t.$$

For each simulation run, the generator enforces that the empirical covariance $\hat{\Sigma} = \frac{1}{T-1} \sum_t (w_t - \bar{w})(w_t - \bar{w})^\top$ belongs to the ambiguity ball $\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma)$. If a custom covariance Σ is supplied, it is first projected onto the ball via the Bures geodesic contraction to guarantee membership.

Purpose and benefits. This sampling scheme serves two purposes:

- It ensures that simulated disturbances remain consistent with the uncertainty model used in the distributionally robust LMI problem.
- It provides a tunable mechanism to test robustness against either uncorrelated (iid) or correlated noise realizations, while maintaining the same nominal covariance structure Σ_{nom}.
- The parameter γ defines the radius of admissible covariance deviation: larger γ values enlarge the ambiguity set and correspond to stronger robustness guarantees.

In summary, the disturbances in simulation are not arbitrary noise sequences but samples from a *Wasserstein-consistent* uncertainty model. This maintains coherence between the data-driven identification, the LMI synthesis, and the subsequent Monte Carlo evaluation of closed-loop robustness.

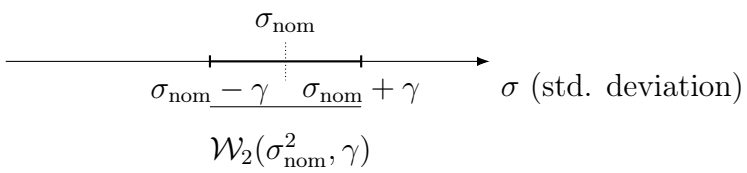

Figure 3.3: 1D Wasserstein ball of zero-mean Gaussians: interval in standard deviation σ.

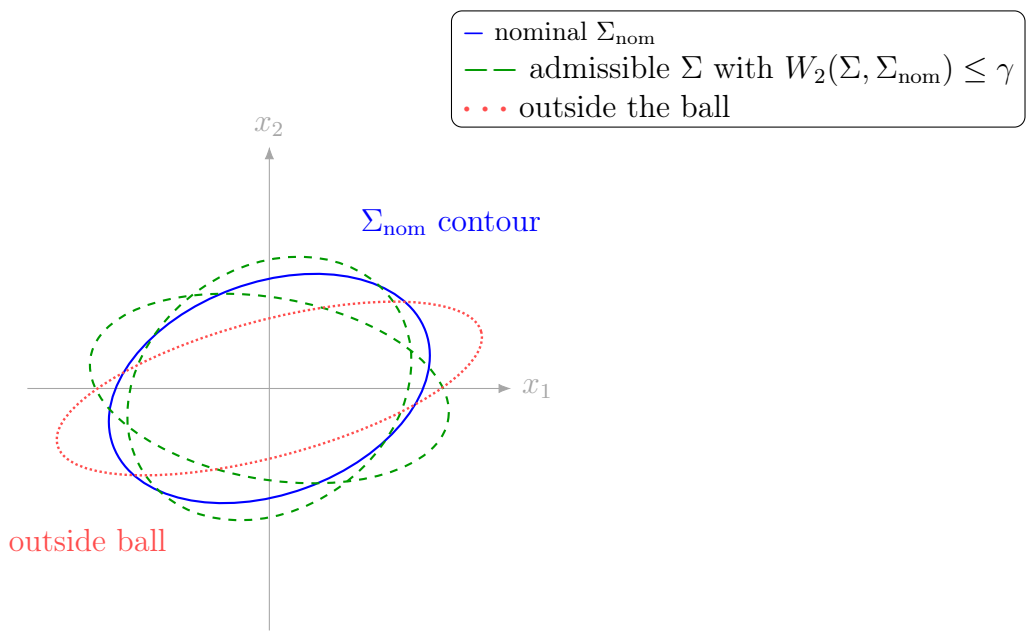

Figure 3.4: 2D zero-mean Gaussians: representative level sets within the W_2 ball around Σ_{nom}.

Chapter 4

Simulation Results

This chapter documents the closed-loop results comparing the *Model-Based Design (MBD)* and *Data-Driven Design (DDD)* pipelines. We report the LMI objective values, a summary figure, controller and plant matrices for both pipelines, and norm-based deltas.

4.1 Objective Values and Summary Figure

- **LMI MBD objective:** 878.5714094275286
- **LMI DDD objective:** 318.34498631060114

4.2 Controller Matrices

MBD Controller

$$A_c^{(\mathrm{MBD})} = \begin{bmatrix} 0.0742989 & 0.7257663 & 0.5497474 & -0.0950312 & -0.0455852 & -0.0091134 & 0.2362168 \\ 0.0070979 & 0.8638508 & -0.0516011 & -0.0720109 & -0.0222940 & 0.0031792 & -0.0376084 \\ 0.1272601 & -0.0964721 & 0.8679311 & 0.1308790 & 0.0362048 & -0.0191056 & 0.0037334 \\ 0.0250123 & -0.0344312 & 0.0509930 & 0.0846822 & 0.0266042 & -0.1597391 & -0.0966628 \\ 0.0144273 & -0.0229488 & 0.0017147 & -0.0300213 & 0.0068849 & 0.4207096 & -0.0874741 \\ 0.0006611 & -0.0009258 & -0.0032466 & 0.0220548 & -0.0228075 & -0.0137108 & -0.0052132 \\ -0.1403130 & 0.2191793 & 0.2020515 & -0.3711880 & -0.1597152 & 0.0273305 & 0.9098114 \end{bmatrix}$$

$$B_c^{(\mathrm{MBD})} = \begin{bmatrix} 64.6388 & -29.6077 & -0.2706 \\ 453.2805 & 249.0883 & 2.2390 \\ 643.9274 & -389.0403 & -17.9139 \\ 320.8596 & -191.0413 & 599.5163 \\ 231.2342 & -40.5761 & -1717.1514 \\ 13.2228 & 2.8731 & 60.1628 \\ -2424.8795 & -73.3255 & -53.9228 \end{bmatrix},$$

$$C_c^{(\mathrm{MBD})} = \begin{bmatrix} -3.0964\times10^{-3} & 4.6805\times10^{-3} & 3.3328\times10^{-3} & -6.2565\times10^{-4} & -2.9542\times10^{-4} & -4.0287\times10^{-5} \end{bmatrix}$$

$$D_c^{(\mathrm{MBD})} = \begin{bmatrix} 0.29741294 & 0.00070928 & -0.00116277 \end{bmatrix}.$$

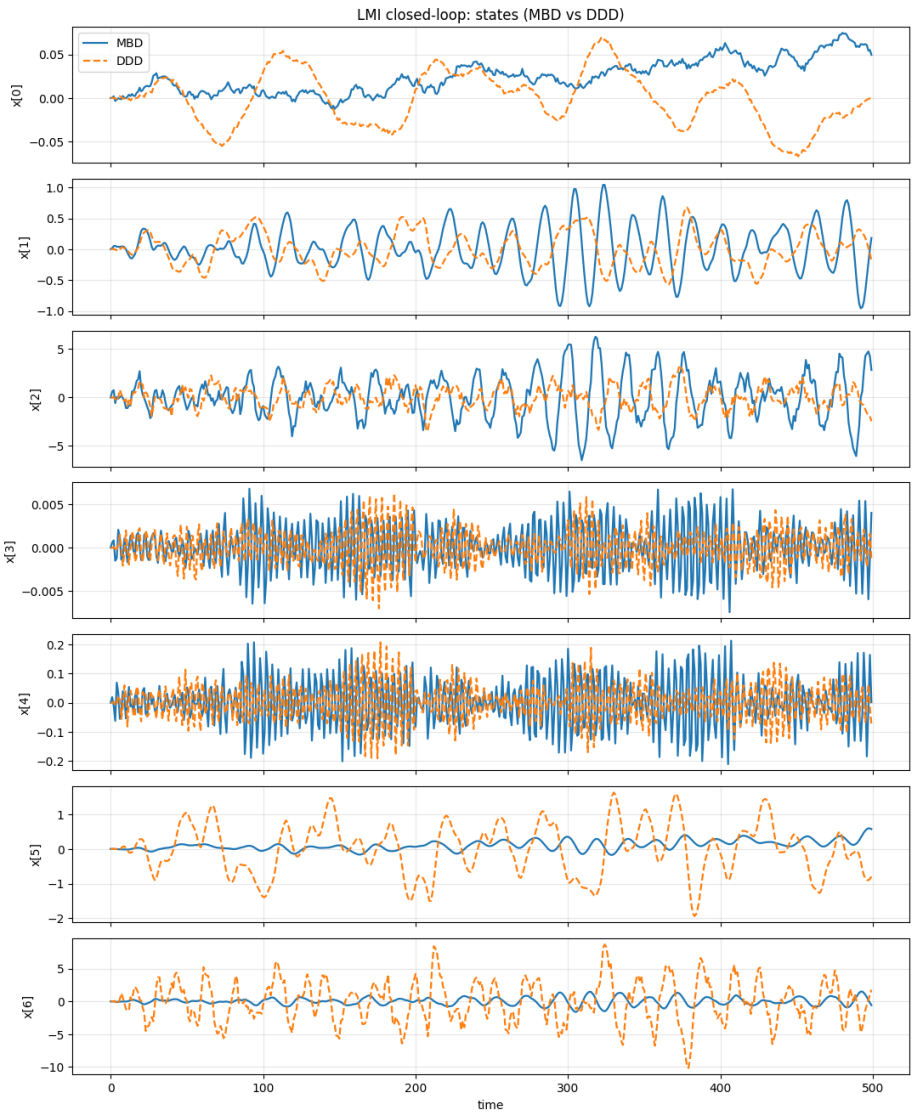

Figure 4.1: Comparison of LMI objectives and closed-loop performance indicators for MBD vs DDD.

DDD Controller

$$A_c^{(\mathrm{DDD})} = \begin{bmatrix} 0.1248681 & 0.2678478 & -0.2918846 & 0.0704319 & 0.0533368 & -0.4278133 & 0.0470614 \\ 0.2658334 & 0.6811754 & 0.2846829 & -0.3017569 & -0.0678523 & -0.5886835 & 0.0576904 \\ 0.2711857 & -0.2325848 & 0.4855597 & -0.0186586 & -0.0102418 & 0.8483247 & -0.2586805 \\ 0.1487607 & -0.4910089 & -1.9795555 & 1.8309049 & 0.2637892 & 5.0586566 & -1.1192087 \\ -0.4231707 & 0.9754647 & 2.6753317 & -1.5373255 & 0.3588473 & -7.5436798 & 1.6831599 \\ 0.0029961 & 0.1064059 & 0.3158765 & -0.2571085 & -0.0273556 & -0.1259121 & 0.0007507 \\ -0.0376768 & -0.1175578 & -1.4844626 & 1.4927367 & 0.1530810 & 4.1381815 & -0.9125240 \end{bmatrix}$$

$$B_c^{(\mathrm{DDD})} = \begin{bmatrix} 246.8528 & -395.9726 & -21.6697 \\ 399.5852 & 377.9486 & 335.6970 \\ -820.0216 & 529.3343 & 452.2322 \\ -808.0278 & -388.0192 & 115.4808 \\ -3163.4195 & -553.3095 & -427.7181 \\ -70.5750 & 332.8711 & 100.0415 \\ -566.5154 & -39.6945 & 19.1882 \end{bmatrix}, \quad C_c^{(\mathrm{DDD})} = \begin{bmatrix} -0.0019110 & 0.0018796 & -0.0013076 & 0.000 \end{bmatrix}$$

$$D_c^{(\mathrm{DDD})} = \begin{bmatrix} 1.2133759 & -0.6348403 & 0.3651989 \end{bmatrix}.$$

4.3 Plant Matrices

MBD Plant

$$A^{(\mathrm{MBD})} = \begin{bmatrix} 9.99875008\times10^{-1} & 1.22964189\times10^{-5} & 3.06714531\times10^{-7} & -8.02852627\times10^{-6} & -2.48360389\times10^{-7} & -1.20633524\times10^{-} \\ 1.22685813\times10^{-4} & 9.51612651\times10^{-1} & 4.84182760\times10^{-2} & -4.54741670\times10^{-10} & -5.86829207\times10^{-12} & -1.20457748\times10^{-} \\ 4.84152089\times10^{-3} & -1.90953931\times10^{0} & 9.21205974\times10^{-1} & -2.48855519\times10^{-8} & -4.47371095\times10^{-10} & -4.67006807\times10^{-} \\ 9.93441555\times10^{-7} & 4.51056383\times10^{-12} & 5.86829207\times10^{-14} & 2.04616572\times10^{-2} & 3.08688261\times10^{-2} & -9.73032487\times10^{-} \\ 3.08663425\times10^{-5} & 2.46046028\times10^{-10} & 4.47371095\times10^{-12} & -3.04354278\times10^{1} & -1.83095883\times10^{-2} & -2.93660233\times10^{-} \\ 0 & 0 & 0 & 0 & 0 & 9.01672541\times10^{-} \\ 0 & 0 & 0 & 0 & 0 & -3.44904321\times10^{0} \end{bmatrix}$$

$$B_w^{(\mathrm{MBD})} = \begin{bmatrix} 2.79495425\times10^{-3} & 4.39653430\times10^{-8} & 1.0\times10^{-4} & 0 & \cdots & 0 \\ 1.09739276\times10^{-2} & 1.09738126\times10^{-2} & 0 & 1.0\times10^{-4} & \cdots & 0 \\ 4.33073084\times10^{-1} & 4.33066226\times10^{-1} & 0 & 0 & 1.0\times10^{-4} & 0 \\ 4.44301862\times10^{-4} & 4.44300848\times10^{-4} & 0 & 0 & 0 & 1.0\times10^{-4} \\ 1.38050142\times10^{-2} & 1.38049587\times10^{-2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0\times10^{-4} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \ (7\times12), \quad B_u^{(\mathrm{MBD})} = \begin{bmatrix} -4.3592\times10^{-5} \\ -2.2383\times10^{-5} \\ -1.7207\times10^{-3} \\ -2.0500\times10^{-7} \\ -1.5053\times10^{-5} \\ 9.8327\times10^{-2} \\ 3.4490\times10^{0} \end{bmatrix}$$

$$C_z^{(\mathrm{MBD})} = \begin{bmatrix} I_7 \\ \mathbf{0}_{1\times7} \end{bmatrix}, \quad D_{zw}^{(\mathrm{MBD})} = \mathbf{0}_{8\times12}, \quad D_{zu}^{(\mathrm{MBD})} = \begin{bmatrix} \mathbf{0}_{7\times1} \\ 1 \end{bmatrix},$$

$$C_y^{(\mathrm{MBD})} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad D_{yw}^{(\mathrm{MBD})} = \mathrm{diag}(0,0,0,0,0,0,0,10^{-4},10^{-4},10^{-4})_{3\times12}.$$

DDD Plant

$$A^{(\mathrm{DDD})} = \begin{bmatrix} 1.00020468 & 1.42244868\times10^{-3} & 9.97365710\times10^{-5} & 3.33898196\times10^{-2} & -6.81374983\times10^{-6} & -1.14552780\times10^{-3} & 2. \\ 2.77052295\times10^{-3} & 0.96146064 & 4.90904756\times10^{-2} & 2.60738710\times10^{-1} & 8.86575616\times10^{-4} & -2.12169849\times10^{-3} & 4. \\ 1.09334995\times10^{-1} & -1.52090109 & 0.94773348 & 1.02897175\times10^{1} & 3.49874659\times10^{-2} & -8.28935343\times10^{-2} & 1. \\ 1.08196988\times10^{-4} & 3.98717336\times10^{-4} & 2.72154752\times10^{-5} & 3.10180258\times10^{-2} & 3.09047206\times10^{-2} & -4.68622227\times10^{-5} & 1. \\ 3.36179952\times10^{-3} & 1.23885003\times10^{-2} & 8.45607213\times10^{-4} & -3.01071052\times10^{1} & -1.71939807\times10^{-2} & -1.44740275\times10^{-3} & 5. \\ -4.2358\times10^{-11} & -3.5895\times10^{-12} & -1.8440\times10^{-11} & 1.8475\times10^{-11} & -8.5631\times10^{-11} & 0.90167253 & \\ 1.0021\times10^{-9} & -1.0292\times10^{-9} & 1.3862\times10^{-10} & -1.1795\times10^{-8} & 7.8009\times10^{-10} & -3.44904312 & \end{bmatrix}$$

$$B_w^{(\mathrm{DDD})} = \begin{bmatrix} -1.7116\times10^{-3} & 1.0\times10^{-4} & 0 & \cdots & 0 \\ -1.3815\times10^{-2} & 0 & 1.0\times10^{-4} & \cdots & 0 \\ -5.4518\times10^{-1} & 0 & 0 & 1.0\times10^{-4} & 0 \\ -5.5932\times10^{-4} & 0 & 0 & 0 & 1.0\times10^{-4} \\ -1.7379\times10^{-2} & 0 & 0 & 0 & 0 \\ -1.3155\times10^{-18} & 0 & 0 & 0 & 0 \\ -5.4917\times10^{-17} & 0 & 0 & 1.0\times10^{-4} & 0 \end{bmatrix} \ (7\times11), \quad B_u^{(\mathrm{DDD})} = \begin{bmatrix} -1.7213\times10^{-3} \\ -6.1172\times10^{-3} \\ -2.4225\times10^{-1} \\ -2.4697\times10^{-4} \\ -7.6821\times10^{-3} \\ 9.8327\times10^{-2} \\ 3.4490\times10^{0} \end{bmatrix}$$

$$C_z^{(\mathrm{DDD})} \approx I_7 \ (\text{with } <10^{-8} \text{ perturbations}), \qquad D_{zw}^{(\mathrm{DDD})} \approx \mathbf{0}_{8\times11}, \quad D_{zu}^{(\mathrm{DDD})} \approx \begin{bmatrix} \mathbf{0}_{7\times1} \\ 1 \end{bmatrix},$$

$$C_y^{(\mathrm{DDD})} \approx \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \qquad D_{yw}^{(\mathrm{DDD})} = \text{selector with } 10^{-4} \text{ on three channels.}$$

4.4 Delta Summaries (DDD minus MBD)

Controller deltas

$\|\Delta A_c\|_F = 1.195\times10^{1}$, $\|\Delta B_c\|_F = 4.699\times10^{3}$, $\|\Delta C_c\|_F = 6.237\times10^{-3}$, $\|\Delta D_c\|_F = 1.174$.
$\max|\Delta A_c| = 7.964$, $\max|\Delta B_c| = 3.395\times10^{3}$, $\max|\Delta C_c| = 4.640\times10^{-3}$, $\max|\Delta D_c| = 9.160\times10^{-1}$.

Plant deltas

$\|\Delta A\|_F = 1.031\times10^{1}$, $\|\Delta B_u\|_F = 2.407\times10^{-1}$, $\|\Delta C_y\|_F = 2.000$, $\|\Delta C_z\|_F = 1.050\times10^{-5}$, $\|\Delta D_z$
$\max|\Delta A| = 1.029\times10^{1}$, $\max|\Delta B_u| = 2.405\times10^{-1}$, $\max|\Delta C_y| = 1.000$, $\max|\Delta C_z| = 1.050\times10^{-5}$,

4.5 Interpretation

Current evidence suggests the DDD pipeline achieves a substantially smaller LMI objective under the same uncertainty model, with controller gains that are larger in magnitude, especially in B_c. The plant deltas indicate modest shifts in A and B_u (relative to scale), while C_z and D_{zu} remain effectively unchanged. These differences align with a controller that trades stronger actuation and feedthrough for improved robust performance under the Wasserstein ambiguity set. Further statistical validation (held-out trajectories, confidence intervals on costs) would be required to claim significance beyond point estimates.

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